

xtime: Monte Carlo Simulation Results

Interactive Fixed Effects Estimators — Bai (2009), Moon & Weidner (2017), SWW (2025)

$B = 1000$ replications per cell; $r_{\text{true}} = 2$; $\beta_{\text{true}} = 1$ (static) / 0.5 (dynamic)

June 12, 2026

Data generating processes

We report simulation results based on $B = 1000$ replications of the panel model

$$y_{it} = \beta x_{it} + \sum_{\rho=1}^2 \lambda_{i\rho} f_{t\rho} + u_{it}, \quad i = 1, \dots, N, \quad t = 1, \dots, T,$$

with $r_{\text{true}} = 2$ interactive factors and no additive fixed effects ($\alpha_i = \xi_t = 0$). Throughout, the loadings are $\lambda_{i\rho} \stackrel{iid}{\sim} N(0, 1)$, independent across i and ρ and of all other primitives. The following four DGPs are employed.

DGP 1: static balanced panel. We set $\beta_0 = 1$ and, for $\rho \in \{1, 2\}$, $f_{t\rho} \stackrel{iid}{\sim} N(0, 1)$ and $\mu_{i\rho} \stackrel{iid}{\sim} N(0, 1)$, with μ_i independent of λ_i . The regressor is

$$x_{it} = \sum_{\rho=1}^2 (\lambda_{i\rho} + \mu_{i\rho}) f_{t\rho} + \varepsilon_{it}, \quad \varepsilon_{it} \stackrel{iid}{\sim} N(0, 1),$$

so x_{it} is strictly exogenous with respect to u_{it} but loads on the same factors as y_{it} : pooled OLS that ignores the factor structure is biased (the OLS column, ≈ 0.4). The independent extra loading μ_i ensures x is not an exact linear combination of (f_{t1}, f_{t2}) ; otherwise the projection of y on x would absorb one factor direction and the information criterion would select $\hat{r} = 1$. Errors u_{it} follow the homoskedastic or heteroskedastic specification described below. The panel is fully observed.

DGP 2: static unbalanced panel. Identical to DGP 1, after which cells are removed completely at random: a subset of $\lfloor \pi NT \rfloor$ index pairs (i, t) is drawn uniformly without replacement and retained, $\pi \in \{0.8, 0.6\}$ (the “fill” rate). The missingness indicator d_{it} is therefore independent of $(\lambda_i, f_t, u_{is}, x_{is})$ for all (i, s) — SWW’s missing Pattern 1 with a homogeneous observation probability.

DGP 3: dynamic balanced panel. We set $\beta_0 = 0.5$ and let the factors follow stationary AR(1) processes,

$$f_{t\rho} = 0.5 f_{t-1,\rho} + \eta_{t\rho}, \quad \eta_{t\rho} \stackrel{iid}{\sim} N(0, 1 - 0.5^2), \quad f_{1\rho} \sim N(0, 1),$$

so that $\text{Var}(f_{t\rho}) = 1$ for all t . The outcome is generated recursively over $t = 1, \dots, T_0 + T$ with burn-in length $T_0 = 50$:

$$y_{i1} = \lambda_i' f_1 + u_{i1}, \quad y_{it} = 0.5 y_{i,t-1} + \lambda_i' f_t + u_{it}, \quad t \geq 2,$$

and the regressor is the lagged outcome, $x_{it} = y_{i,t-1}$. The first $T_0 + 1$ periods are discarded, leaving an estimation sample of $T - 1$ time periods (each retained y_{it} paired with its observed lag). The errors u_{it} are *serially uncorrelated* in both variants: with a lagged dependent variable, serial correlation in u_{it} would render $y_{i,t-1}$ contemporaneously endogenous ($\mathbb{E}[x_{it}u_{it}] \neq 0$), which no incidental-parameter bias correction can repair. Thus x_{it} is predetermined (weakly exogenous): $\mathbb{E}[u_{it} | y_{i,t-1}, y_{i,t-2}, \dots, \lambda_i, f] = 0$. This design generates the Nickell-type $O(1/T)$ incidental-parameter bias, which halves as T doubles (visible in the IFE column of Table 3).

DGP 4: dynamic unbalanced panel. Identical to DGP 3, after which cells are removed completely at random exactly as in DGP 2 ($\lfloor \pi N(T - 1) \rfloor$ cells retained, $\pi \in \{0.8, 0.6\}$). Missing lags are handled by the unbalanced estimator's observed-cell objective; no imputation of x is performed.

Error variants. In the *homoskedastic* tables, $u_{it} \stackrel{iid}{\sim} N(0, 1)$. In the *heteroskedastic* tables, $u_{it} = \sigma_{it} e_{it}$ with $e_{it} \stackrel{iid}{\sim} N(0, 1)$ and

$$\sigma_{it} = c \left(0.4 + 0.8 \frac{|\lambda'_i f_t|}{\text{sd}(\lambda' f)} \right),$$

where $\text{sd}(\lambda' f)$ is the in-sample standard deviation of the factor component and the constant c rescales σ_{it} so that $\frac{1}{NT} \sum_{it} \sigma_{it}^2 = 1$, keeping the overall noise level identical to the homoskedastic case. Because σ_{it} rises with $|\lambda'_i f_t|$ — the same component the regressor loads on — the homoskedastic standard SE undercovers by design, and valid inference requires the robust/cluster/HAC variants.

Estimation. For every replication we compute: (i) pooled OLS ignoring the factor structure (grand-mean demeaned; for the unbalanced designs, OLS with intercept on the observed cells); (ii) the raw IFE estimator with $r = 2$ treated as known — `ife()` for DGPs 1 and 3 (the Bai 2009 least-squares estimator; `method = "dynamic"` for DGP 3) and `ife_unbalanced()` for DGPs 2 and 4 (the SWW 2025 EM estimator with nuclear-norm-regularised initialisation, `init = "nnr"`); and (iii) the bias-corrected estimator: Bai (2009) $B/N + C/T$ for DGP 1; SWW b_3 – b_6 under strict exogeneity for DGP 2; Moon–Weidner (2017) B_1, B_2, B_3 with lag bandwidth $M_1 = 1$ for DGP 3; and the SWW weak-exogeneity correction b_2 – b_6 for DGP 4, where the dynamic term b_2 and the HAC variance use a Bartlett kernel with bandwidth $L_T = \lfloor 2T^{1/5} \rfloor$. Confidence intervals for the raw estimator are reported with homoskedastic standard, HC1-robust, cluster-robust (by unit), and — for the unbalanced estimator — HAC standard errors; the bias-corrected estimator is paired with the SE matching the error structure (standard under homoskedasticity, HC1-robust under heteroskedasticity, HAC for DGP 4). The number of factors is selected by a BIC-type information criterion over $r \in \{0, 1, \dots, 5\}$; selection accuracy is reported as $P(\hat{r} = r)$ in Tables 1–4 and in full in Table 5. Replications use the L'Ecuyer–CMRG generator with a deterministic seed per (DGP, N, T, π, b) cell, so every entry is exactly reproducible.

Grid. $N \in \{30, 50, 100, 200\}$, $T \in \{10, 50, 100, 200\}$ ($T = 10$ omitted for the dynamic designs, which require the burn-in), and fill $\pi \in \{80\%, 60\%\}$ for DGPs 2 and 4. Note that the loadings are mean-zero and the missingness non-informative, so the intercept-free specification (`force = "none"`) is correctly specified throughout; this deliberately differs from the simulation design of SWW (2025), which uses mean-one loadings, two regressors, and factor-correlated missing patterns.

Notation and definitions

Each cell of Tables 1–4 summarises $B = 1000$ Monte Carlo replications. Let $\hat{\beta}_b$ denote the estimate in replication $b = 1, \dots, B$, β_0 the true coefficient, $\bar{\hat{\beta}} = B^{-1} \sum_b \hat{\beta}_b$, and $\widehat{\text{se}}_b$ the estimated standard error in replication b . The reported statistics are:

- **Bias** (columns OLS, IFE, BC): $\text{Bias} = B^{-1} \sum_{b=1}^B (\hat{\beta}_b - \beta_0)$, computed separately for pooled OLS (ignoring the factor structure), the raw IFE estimator with r treated as known, and the analytically bias-corrected (BC) IFE estimator.
- **SD**: Monte Carlo standard deviation of the raw IFE estimator, $\text{SD} = [(B - 1)^{-1} \sum_b (\hat{\beta}_b - \bar{\hat{\beta}})^2]^{1/2}$.
- **RMSE / BC.RMSE**: $\text{RMSE} = [B^{-1} \sum_b (\hat{\beta}_b - \beta_0)^2]^{1/2} = \sqrt{\text{Bias}^2 + \frac{B-1}{B} \text{SD}^2}$, for the raw and the bias-corrected estimator respectively.

- **SE/SD**: calibration ratio $B^{-1} \sum_b \widehat{\text{se}}_b / \text{SD}$ using the homoskedastic standard SE of the raw estimator. Values near 1 indicate that the analytical SE formula replicates the true sampling variability; values below 1 indicate underestimation (expected for the *standard* SE under heteroskedastic errors).
- **95% Coverage** (Std / Rob / Cl / HAC): empirical coverage of the nominal 95% confidence interval for the *raw* estimator, $B^{-1} \sum_b \mathbf{1}\{|\hat{\beta}_b - \beta_0| \leq 1.96 \widehat{\text{se}}_b\}$, where $\widehat{\text{se}}_b$ is, respectively, the homoskedastic standard SE, the HC1 heteroskedasticity-robust SE, the cluster-robust SE (clustered by unit), or the Bartlett-kernel HAC SE with bandwidth $L_T = \lfloor 2T^{1/5} \rfloor$ (unbalanced estimator only).
- **BC.Std / BC.Rob / BC.HAC**: coverage of the same nominal 95% interval centred at the *bias-corrected* estimator, paired with the SE matching the error structure: standard under homoskedastic errors, HC1-robust under heteroskedastic errors, HAC for the dynamic unbalanced design (DGP 4).
- **Size**: empirical rejection rate of the two-sided t -test of $H_0: \beta = \beta_0$ at the 5% nominal level using the raw estimator with the standard SE, $B^{-1} \sum_b \mathbf{1}\{|\hat{\beta}_b - \beta_0| / \widehat{\text{se}}_b > 1.96\}$. Size distortions mirror the bias of the raw estimator ($\text{bias}/\text{SE} \sim \sqrt{N/T}$ in the dynamic designs).
- $P(\hat{r} = r)$: percentage of replications in which the information criterion (BIC-type, searched over $r = 0, \dots, 5$) selects the true number of factors $r = 2$.
- **Conv.**: fraction of replications in which the alternating estimation loop converged within the iteration limit.

Table 5 reports the factor-selection distribution: $\bar{r}$ is the mean selected factor number, and $P(\hat{r} = 2)$, $P(\hat{r} < 2)$, $P(\hat{r} > 2)$ are the percentages of replications with correct selection, underselection, and overselection.

1 Homoskedastic errors

Table 1: Monte Carlo results: static balanced panel (DGP 1). N : units; T : periods; OLS/IFE/BC: bias of each estimator; SE/SD: ratio of mean SE to Monte Carlo SD; Cov: empirical 95% coverage; Size: rejection rate under H_0 .

N	T	Bias & RMSE						95% Coverage							
		OLS	IFE	SD	RMSE	BC	BC.RMSE	SE/SD	Std	Rob	CI	BC.Std	Size	$P(\hat{r} = r)$	Conv.
30	10	0.394	0.021	0.076	0.079	0.021	0.079	0.773	.872	.814	.843	.865	.133	0.1	1.000
30	50	0.396	0.004	0.029	0.029	0.004	0.029	0.846	.896	.880	.870	.898	.104	43.4	1.000
30	100	0.397	0.002	0.020	0.020	0.002	0.020	0.855	.910	.888	.873	.910	.091	75.8	1.000
30	200	0.400	0.001	0.014	0.015	0.001	0.015	0.840	.907	.894	.885	.908	.093	83.9	1.000
50	10	0.394	0.010	0.056	0.057	0.010	0.057	0.845	.900	.857	.878	.900	.100	0.1	1.000
50	50	0.398	0.002	0.022	0.022	0.002	0.022	0.909	.920	.912	.916	.919	.080	93.8	1.000
50	100	0.400	0.001	0.015	0.015	0.001	0.015	0.906	.927	.918	.915	.925	.073	100.0	1.000
50	200	0.399	0.001	0.010	0.010	0.001	0.010	0.965	.946	.936	.926	.944	.054	100.0	1.000
100	10	0.392	0.004	0.037	0.037	0.004	0.037	0.934	.934	.903	.925	.935	.066	0.0	1.000
100	50	0.398	0.001	0.015	0.015	0.001	0.015	0.970	.945	.938	.939	.946	.055	99.9	1.000
100	100	0.399	0.001	0.010	0.010	0.001	0.011	0.944	.937	.930	.924	.935	.063	100.0	1.000
100	200	0.398	0.000	0.007	0.007	0.000	0.007	0.964	.934	.928	.929	.934	.066	100.0	1.000
200	10	0.391	0.002	0.026	0.026	0.002	0.026	0.959	.942	.911	.940	.943	.058	0.0	1.000
200	50	0.397	0.000	0.010	0.010	0.000	0.010	0.998	.954	.944	.949	.955	.046	100.0	1.000
200	100	0.399	-0.000	0.007	0.007	-0.000	0.007	0.978	.949	.942	.942	.947	.051	100.0	1.000
200	200	0.401	0.000	0.005	0.005	0.000	0.005	0.950	.936	.935	.933	.937	.064	100.0	1.000

Table 2: Monte Carlo results: static unbalanced panel (DGP 2, fill = 80%).

N	T	Bias & RMSE						95% Coverage							
		OLS	IFE	SD	RMSE	BC	BC.RMSE	SE/SD	Std	Rob	CI	BC.Std	Size	$P(\hat{r} = r)$	Conv.
30	10	0.385	0.042	0.108	0.116	0.043	0.117	0.644	.809	.740	.780	.802	.197	61.7	0.974
30	50	0.397	0.005	0.032	0.033	0.005	0.033	0.916	.923	.901	.887	.926	.078	85.7	1.000
30	100	0.398	0.002	0.023	0.023	0.002	0.023	0.900	.913	.898	.892	.913	.087	92.0	1.000
30	200	0.396	0.001	0.016	0.016	0.001	0.016	0.895	.923	.903	.903	.914	.077	93.1	1.000
50	10	0.392	0.018	0.070	0.072	0.019	0.073	0.806	.878	.825	.867	.878	.123	75.5	0.991
50	50	0.396	0.001	0.025	0.025	0.001	0.025	0.922	.924	.911	.918	.923	.076	96.1	1.000
50	100	0.398	0.001	0.017	0.017	0.001	0.017	0.943	.937	.928	.929	.938	.063	98.7	1.000
50	200	0.398	0.000	0.012	0.012	0.000	0.012	0.917	.929	.913	.914	.926	.071	99.3	1.000
100	10	0.394	0.005	0.043	0.043	0.005	0.043	0.945	.936	.898	.929	.934	.065	81.1	1.000
100	50	0.396	0.001	0.017	0.017	0.001	0.017	0.965	.939	.926	.930	.938	.061	98.5	1.000
100	100	0.397	-0.000	0.012	0.012	-0.000	0.012	0.942	.935	.927	.931	.937	.065	99.9	1.000
100	200	0.398	0.000	0.008	0.008	0.000	0.008	0.985	.941	.938	.936	.943	.059	99.9	1.000
200	10	0.394	0.003	0.030	0.030	0.003	0.030	0.965	.938	.905	.940	.937	.062	85.6	1.000
200	50	0.399	0.000	0.012	0.012	0.000	0.012	0.968	.947	.939	.945	.946	.053	99.2	1.000
200	100	0.399	-0.000	0.009	0.009	-0.000	0.009	0.928	.923	.919	.919	.922	.077	100.0	1.000
200	200	0.399	-0.000	0.006	0.006	-0.000	0.006	0.955	.937	.935	.938	.937	.063	100.0	1.000

Table 3: Monte Carlo results: static unbalanced panel (DGP 2, fill = 60%).

		Bias & RMSE							95% Coverage						
N	T	OLS	IFE	SD	RMSE	BC	BC.RMSE	SE/SD	Std	Rob	Cl	BC.Std	Size	$P(\hat{r} = r)$	Conv.
30	10	0.384	0.130	0.157	0.204	0.131	0.205	0.475	.536	.461	.512	.529	.471	57.8	0.813
30	50	0.396	0.008	0.041	0.042	0.008	0.042	0.826	.889	.853	.854	.885	.111	76.9	1.000
30	100	0.398	0.005	0.028	0.028	0.005	0.028	0.854	.902	.876	.871	.900	.098	89.3	1.000
30	200	0.396	0.002	0.020	0.020	0.002	0.020	0.846	.896	.885	.869	.895	.104	91.1	1.000
50	10	0.393	0.063	0.113	0.130	0.065	0.131	0.569	.703	.627	.690	.699	.301	67.6	0.863
50	50	0.397	0.002	0.030	0.030	0.002	0.030	0.896	.917	.895	.892	.913	.085	90.0	1.000
50	100	0.398	0.002	0.020	0.020	0.002	0.020	0.916	.935	.913	.908	.932	.065	95.9	1.000
50	200	0.398	0.001	0.014	0.014	0.001	0.014	0.915	.930	.916	.912	.928	.070	97.8	1.000
100	10	0.395	0.026	0.059	0.065	0.027	0.065	0.813	.876	.805	.862	.865	.131	70.2	0.922
100	50	0.396	0.002	0.019	0.020	0.002	0.020	0.968	.937	.928	.929	.937	.063	96.4	1.000
100	100	0.396	0.000	0.014	0.014	0.000	0.014	0.919	.928	.920	.924	.928	.072	99.8	1.000
100	200	0.398	0.000	0.009	0.009	0.000	0.009	0.974	.947	.940	.934	.944	.053	100.0	1.000
200	10	0.394	0.010	0.036	0.037	0.010	0.037	0.962	.949	.861	.924	.937	.051	79.7	0.987
200	50	0.399	0.001	0.014	0.014	0.001	0.014	0.966	.944	.928	.936	.944	.056	99.1	1.000
200	100	0.399	-0.000	0.010	0.010	-0.000	0.010	0.939	.936	.932	.930	.936	.064	99.9	1.000
200	200	0.398	0.000	0.007	0.007	0.000	0.007	0.943	.940	.935	.935	.938	.060	100.0	1.000

Table 4: Monte Carlo results: dynamic balanced panel (DGP 3). Regressors: $x_{it} = y_{i,t-1}$; bias correction via Moon & Weidner (2017). $T \geq 50$ only (burn-in requirement). *Cluster-robust SE used as within-unit correlation correction (`ife()` has no explicit HAC option).

		Bias & RMSE							95% Coverage						
N	T	OLS	IFE	SD	RMSE	BC	BC.RMSE	SE/SD	Std	Rob	Cl*	BC.Std	Size	$P(\hat{r} = r)$	Conv.
30	50	0.220	-0.017	0.029	0.033	-0.002	0.027	0.859	.854	.835	.837	.936	.147	82.9	1.000
30	100	0.227	-0.009	0.018	0.020	-0.001	0.018	0.922	.892	.880	.875	.935	.108	95.4	1.000
30	200	0.228	-0.003	0.013	0.013	0.000	0.013	0.928	.923	.913	.895	.932	.077	98.0	1.000
50	50	0.220	-0.020	0.021	0.029	-0.005	0.021	0.890	.802	.771	.799	.927	.199	98.1	1.000
50	100	0.225	-0.010	0.014	0.017	-0.002	0.013	0.952	.872	.859	.861	.945	.128	99.9	1.000
50	200	0.227	-0.005	0.009	0.011	-0.001	0.009	0.964	.918	.899	.903	.942	.082	100.0	1.000
100	50	0.220	-0.020	0.016	0.026	-0.005	0.016	0.823	.653	.641	.661	.914	.347	99.8	1.000
100	100	0.224	-0.010	0.010	0.014	-0.002	0.010	0.918	.800	.792	.796	.929	.200	100.0	1.000
100	200	0.228	-0.005	0.006	0.008	-0.001	0.006	0.975	.888	.881	.882	.943	.112	100.0	1.000
200	50	0.221	-0.021	0.012	0.024	-0.006	0.012	0.799	.424	.413	.424	.885	.576	100.0	1.000
200	100	0.227	-0.010	0.007	0.012	-0.002	0.007	0.927	.656	.646	.650	.926	.344	100.0	1.000
200	200	0.228	-0.005	0.005	0.007	-0.001	0.005	0.935	.787	.785	.784	.919	.213	100.0	1.000

Table 5: Monte Carlo results: dynamic unbalanced panel (DGP 4, fill = 80%).

		Bias & RMSE						95% Coverage							
N	T	OLS	IFE	SD	RMSE	BC	BC.RMSE	SE/SD	Std	Rob	Cov.HAC	BC.HAC	Size	$P(\hat{r} = r)$	Conv.
30	50	0.222	-0.020	0.032	0.038	-0.009	0.032	0.874	.860	.834	.833	.897	.140	90.1	1.000
30	100	0.224	-0.011	0.020	0.022	-0.004	0.020	0.968	.913	.890	.886	.929	.087	97.3	1.000
30	200	0.227	-0.005	0.013	0.014	-0.002	0.013	1.003	.938	.926	.927	.945	.062	98.5	1.000
50	50	0.223	-0.021	0.024	0.032	-0.009	0.024	0.902	.812	.790	.791	.898	.188	97.7	1.000
50	100	0.224	-0.010	0.015	0.018	-0.004	0.015	0.966	.885	.871	.868	.927	.115	99.7	1.000
50	200	0.228	-0.005	0.011	0.012	-0.001	0.011	0.950	.911	.892	.895	.931	.089	100.0	1.000
100	50	0.221	-0.022	0.018	0.028	-0.010	0.019	0.834	.665	.647	.645	.855	.335	99.5	1.000
100	100	0.227	-0.010	0.011	0.015	-0.003	0.011	0.934	.818	.806	.810	.928	.182	100.0	1.000
100	200	0.229	-0.005	0.008	0.009	-0.002	0.008	0.941	.869	.863	.863	.922	.131	100.0	1.000
200	50	0.224	-0.022	0.013	0.026	-0.010	0.015	0.802	.451	.434	.438	.812	.549	99.7	1.000
200	100	0.225	-0.010	0.008	0.013	-0.003	0.008	0.942	.707	.693	.692	.919	.293	100.0	1.000
200	200	0.227	-0.005	0.005	0.007	-0.001	0.005	0.972	.843	.834	.836	.945	.157	100.0	1.000

Table 6: Monte Carlo results: dynamic unbalanced panel (DGP 4, fill = 60%).

		Bias & RMSE						95% Coverage							
N	T	OLS	IFE	SD	RMSE	BC	BC.RMSE	SE/SD	Std	Rob	Cov.HAC	BC.HAC	Size	$P(\hat{r} = r)$	Conv.
30	50	0.223	-0.019	0.038	0.043	-0.009	0.038	0.845	.875	.835	.835	.881	.126	87.7	1.000
30	100	0.224	-0.010	0.023	0.025	-0.004	0.023	0.966	.917	.893	.892	.914	.084	97.2	1.000
30	200	0.226	-0.005	0.016	0.017	-0.002	0.016	0.985	.937	.917	.918	.934	.063	99.2	1.000
50	50	0.222	-0.021	0.029	0.035	-0.010	0.029	0.857	.836	.803	.804	.876	.165	95.0	1.000
50	100	0.224	-0.010	0.018	0.020	-0.004	0.018	0.949	.897	.883	.883	.921	.103	99.5	1.000
50	200	0.228	-0.005	0.012	0.013	-0.001	0.012	0.967	.926	.915	.915	.940	.074	100.0	1.000
100	50	0.221	-0.021	0.021	0.030	-0.010	0.022	0.834	.724	.701	.703	.872	.276	98.7	1.000
100	100	0.227	-0.010	0.013	0.016	-0.003	0.013	0.932	.853	.841	.842	.921	.147	100.0	1.000
100	200	0.228	-0.005	0.009	0.010	-0.002	0.009	0.946	.889	.880	.878	.926	.111	100.0	1.000
200	50	0.224	-0.022	0.015	0.026	-0.010	0.017	0.812	.557	.534	.536	.805	.443	100.0	1.000
200	100	0.225	-0.010	0.009	0.013	-0.003	0.009	0.935	.764	.746	.742	.913	.236	100.0	1.000
200	200	0.227	-0.005	0.006	0.008	-0.001	0.006	0.955	.853	.845	.843	.938	.147	100.0	1.000

Table 7: Factor selection accuracy across all DGPs ($r_{\text{true}} = 2$, $r_{\text{max}} = 5$).

DGP	Fill	N	T	$\bar{r}$	$P(\hat{r} = 2)\%$	$P(\hat{r} < 2)\%$	$P(\hat{r} > 2)\%$
1	80%	30	10	1.00	0.1	99.9	0.0
1	80%	30	50	1.43	43.4	56.6	0.0
1	80%	30	100	1.76	75.8	24.2	0.0
1	80%	30	200	1.84	83.9	16.1	0.0
1	80%	50	10	1.00	0.1	99.9	0.0
1	80%	50	50	1.94	93.8	6.2	0.0
1	80%	50	100	2.00	100.0	0.0	0.0
1	80%	50	200	2.00	100.0	0.0	0.0
1	80%	100	10	1.00	0.0	100.0	0.0
1	80%	100	50	2.00	99.9	0.1	0.0
1	80%	100	100	2.00	100.0	0.0	0.0
1	80%	100	200	2.00	100.0	0.0	0.0
1	80%	200	10	1.00	0.0	100.0	0.0
1	80%	200	50	2.00	100.0	0.0	0.0
1	80%	200	100	2.00	100.0	0.0	0.0
1	80%	200	200	2.00	100.0	0.0	0.0
2	60%	30	10	1.69	57.8	36.5	5.7
2	60%	30	50	1.77	76.9	23.1	0.0
2	60%	30	100	1.89	89.3	10.7	0.0
2	60%	30	200	1.91	91.1	8.9	0.0
2	60%	50	10	1.71	67.6	30.8	1.6
2	60%	50	50	1.90	90.0	10.0	0.0
2	60%	50	100	1.96	95.9	4.1	0.0
2	60%	50	200	1.98	97.8	2.2	0.0
2	60%	100	10	1.70	70.2	29.8	0.0
2	60%	100	50	1.96	96.4	3.6	0.0
2	60%	100	100	2.00	99.8	0.2	0.0
2	60%	100	200	2.00	100.0	0.0	0.0
2	60%	200	10	1.80	79.7	20.3	0.0
2	60%	200	50	1.99	99.1	0.9	0.0
2	60%	200	100	2.00	99.9	0.1	0.0
2	60%	200	200	2.00	100.0	0.0	0.0
2	80%	30	10	1.70	61.7	34.1	4.2
2	80%	30	50	1.86	85.7	14.2	0.1
2	80%	30	100	1.93	92.0	7.8	0.2
2	80%	30	200	1.95	93.1	6.5	0.4
2	80%	50	10	1.78	75.5	23.5	1.0
2	80%	50	50	1.97	96.1	3.8	0.1
2	80%	50	100	1.99	98.7	1.3	0.0
2	80%	50	200	2.00	99.3	0.5	0.2
2	80%	100	10	1.81	81.1	18.8	0.1
2	80%	100	50	1.99	98.5	1.5	0.0
2	80%	100	100	2.00	99.9	0.1	0.0
2	80%	100	200	2.00	99.9	0.0	0.1
2	80%	200	10	1.86	85.6	14.4	0.0
2	80%	200	50	1.99	99.2	0.8	0.0
2	80%	200	100	2.00	100.0	0.0	0.0
2	80%	200	200	2.00	100.0	0.0	0.0
3	80%	30	50	1.83	82.9	17.1	0.0
3	80%	30	100	1.95	95.4	4.6	0.0
3	80%	30	200	1.98	98.0	2.0	0.0
3	80%	50	50	1.98	98.1	1.9	0.0
3	80%	50	100	2.00	99.9	0.1	0.0
3	80%	50	200	2.00	100.0	0.0	0.0
3	80%	100	50	2.00	99.8	0.2	0.0
3	80%	100	100	2.00	100.0	0.0	0.0

2 Heteroskedastic errors

Table 8: Monte Carlo results: static balanced panel (DGP 1). N : units; T : periods; OLS/IFE/BC: bias of each estimator; SE/SD: ratio of mean SE to Monte Carlo SD; Cov: empirical 95% coverage; Size: rejection rate under H_0 .

N	T	Bias & RMSE							95% Coverage						
		OLS	IFE	SD	RMSE	BC	BC.RMSE	SE/SD	Std	Rob	CI	BC.Rob	Size	$P(\hat{r} = r)$	Conv.
30	10	0.394	0.038	0.082	0.091	0.041	0.093	0.627	.788	.737	.799	.724	.213	0.6	1.000
30	50	0.396	0.006	0.030	0.030	0.007	0.031	0.769	.856	.864	.898	.862	.144	65.1	1.000
30	100	0.397	0.004	0.021	0.021	0.004	0.021	0.771	.851	.865	.921	.868	.149	87.7	1.000
30	200	0.400	0.003	0.016	0.016	0.003	0.016	0.734	.844	.863	.936	.871	.156	92.6	1.000
50	10	0.395	0.027	0.060	0.066	0.029	0.068	0.700	.814	.778	.825	.763	.187	0.8	1.000
50	50	0.398	0.003	0.022	0.022	0.003	0.022	0.860	.907	.915	.933	.907	.093	97.6	1.000
50	100	0.400	0.002	0.016	0.016	0.002	0.016	0.842	.897	.903	.929	.917	.103	100.0	1.000
50	200	0.399	0.002	0.011	0.011	0.001	0.010	0.869	.901	.920	.960	.930	.099	100.0	1.000
100	10	0.392	0.016	0.040	0.044	0.018	0.045	0.771	.866	.830	.870	.822	.134	0.4	1.000
100	50	0.398	0.002	0.014	0.014	0.002	0.014	0.959	.938	.937	.948	.933	.062	100.0	1.000
100	100	0.399	0.001	0.010	0.010	0.001	0.010	0.929	.925	.929	.943	.933	.075	100.0	1.000
100	200	0.398	0.000	0.007	0.007	0.000	0.007	0.939	.937	.940	.960	.940	.063	100.0	1.000
200	10	0.392	0.013	0.027	0.030	0.014	0.031	0.831	.877	.838	.884	.826	.123	0.0	1.000
200	50	0.397	0.001	0.010	0.010	0.001	0.010	0.989	.947	.943	.950	.948	.053	100.0	1.000
200	100	0.400	0.001	0.007	0.007	0.000	0.007	0.996	.945	.945	.953	.942	.055	100.0	1.000
200	200	0.401	0.000	0.005	0.005	0.000	0.005	0.984	.945	.952	.953	.950	.055	100.0	1.000

Table 9: Monte Carlo results: static unbalanced panel (DGP 2, fill = 80%).

N	T	Bias & RMSE							95% Coverage						
		OLS	IFE	SD	RMSE	BC	BC.RMSE	SE/SD	Std	Rob	CI	BC.Rob	Size	$P(\hat{r} = r)$	Conv.
30	10	0.385	0.061	0.102	0.119	0.066	0.122	0.578	.714	.654	.709	.632	.286	56.2	0.957
30	50	0.396	0.009	0.034	0.035	0.010	0.035	0.817	.887	.857	.879	.842	.113	83.7	1.000
30	100	0.398	0.005	0.024	0.024	0.005	0.024	0.820	.885	.871	.895	.866	.115	87.6	1.000
30	200	0.396	0.004	0.019	0.019	0.004	0.019	0.724	.836	.823	.861	.829	.164	90.2	1.000
50	10	0.393	0.042	0.078	0.089	0.046	0.091	0.624	.767	.707	.765	.683	.236	69.4	0.981
50	50	0.396	0.004	0.024	0.024	0.004	0.025	0.911	.920	.903	.916	.899	.080	94.9	1.000
50	100	0.399	0.003	0.017	0.017	0.003	0.017	0.899	.912	.902	.909	.910	.088	97.2	1.000
50	200	0.398	0.001	0.012	0.013	0.001	0.012	0.869	.911	.899	.917	.907	.089	98.7	1.000
100	10	0.394	0.025	0.051	0.056	0.027	0.058	0.708	.818	.780	.840	.762	.182	76.0	0.993
100	50	0.396	0.003	0.017	0.017	0.003	0.017	0.940	.935	.925	.931	.919	.065	98.3	1.000
100	100	0.397	0.001	0.012	0.012	0.001	0.012	0.923	.931	.920	.929	.924	.069	99.7	1.000
100	200	0.398	0.001	0.008	0.008	0.001	0.008	0.960	.945	.942	.944	.948	.055	99.6	1.000
200	10	0.394	0.018	0.034	0.038	0.019	0.039	0.765	.847	.790	.854	.782	.153	80.9	0.998
200	50	0.399	0.002	0.012	0.012	0.002	0.012	0.965	.945	.932	.943	.924	.055	99.3	1.000
200	100	0.399	0.000	0.008	0.008	0.000	0.008	0.965	.941	.935	.938	.935	.059	100.0	1.000
200	200	0.399	-0.000	0.006	0.006	-0.000	0.006	0.975	.949	.948	.941	.948	.051	100.0	1.000

Table 10: Monte Carlo results: static unbalanced panel (DGP 2, fill = 60%).

		Bias & RMSE							95% Coverage						
N	T	OLS	IFE	SD	RMSE	BC	BC.RMSE	SE/SD	Std	Rob	CI	BC.Rob	Size	$P(\hat{r} = r)$	Conv.
30	10	0.384	0.110	0.128	0.169	0.114	0.172	0.481	.544	.479	.534	.471	.462	54.1	0.745
30	50	0.396	0.018	0.046	0.049	0.020	0.051	0.677	.826	.789	.810	.776	.175	73.5	1.000
30	100	0.398	0.009	0.029	0.031	0.010	0.031	0.751	.837	.810	.847	.797	.163	80.8	1.000
30	200	0.396	0.006	0.023	0.024	0.006	0.024	0.666	.804	.789	.844	.798	.197	80.9	1.000
50	10	0.393	0.083	0.106	0.135	0.089	0.139	0.504	.602	.530	.596	.510	.405	65.4	0.849
50	50	0.397	0.008	0.029	0.030	0.009	0.031	0.847	.892	.868	.883	.856	.108	87.6	1.000
50	100	0.398	0.005	0.021	0.022	0.005	0.022	0.837	.882	.872	.891	.868	.118	93.7	1.000
50	200	0.398	0.003	0.015	0.015	0.003	0.015	0.850	.895	.886	.904	.883	.106	96.2	1.000
100	10	0.395	0.050	0.067	0.084	0.055	0.087	0.612	.688	.642	.695	.610	.312	69.5	0.904
100	50	0.396	0.005	0.019	0.020	0.006	0.020	0.930	.923	.910	.921	.911	.077	94.0	1.000
100	100	0.396	0.002	0.014	0.014	0.002	0.014	0.902	.923	.910	.919	.910	.078	99.9	1.000
100	200	0.398	0.001	0.010	0.010	0.001	0.010	0.936	.935	.930	.941	.931	.065	100.0	1.000
200	10	0.394	0.032	0.042	0.053	0.035	0.055	0.717	.797	.759	.835	.734	.203	77.2	0.975
200	50	0.399	0.003	0.014	0.014	0.003	0.014	0.946	.931	.906	.927	.909	.069	97.0	1.000
200	100	0.399	0.001	0.009	0.010	0.001	0.010	0.957	.939	.933	.939	.927	.061	99.9	1.000
200	200	0.399	0.000	0.007	0.007	0.000	0.007	0.953	.944	.939	.941	.939	.056	100.0	1.000

Table 11: Monte Carlo results: dynamic balanced panel (DGP 3). Regressors: $x_{it} = y_{i,t-1}$; bias correction via Moon & Weidner (2017). $T \geq 50$ only (burn-in requirement). *Cluster-robust SE used as within-unit correlation correction (`ife()` has no explicit HAC option).

		Bias & RMSE							95% Coverage						
N	T	OLS	IFE	SD	RMSE	BC	BC.RMSE	SE/SD	Std	Rob	CI*	BC.Rob	Size	$P(\hat{r} = r)$	Conv.
30	50	0.220	-0.027	0.036	0.045	-0.007	0.035	0.686	.722	.759	.758	.878	.278	90.6	1.000
30	100	0.227	-0.014	0.023	0.027	-0.004	0.023	0.726	.776	.837	.814	.905	.225	98.2	1.000
30	200	0.228	-0.006	0.016	0.017	-0.000	0.016	0.736	.826	.882	.871	.924	.174	98.7	1.000
50	50	0.221	-0.032	0.029	0.043	-0.010	0.029	0.658	.581	.664	.664	.881	.419	99.1	1.000
50	100	0.225	-0.015	0.018	0.023	-0.004	0.018	0.726	.719	.812	.792	.917	.281	100.0	1.000
50	200	0.227	-0.008	0.013	0.015	-0.002	0.013	0.718	.767	.846	.839	.907	.233	100.0	1.000
100	50	0.220	-0.033	0.022	0.040	-0.011	0.022	0.598	.406	.498	.514	.854	.594	100.0	1.000
100	100	0.224	-0.017	0.013	0.021	-0.005	0.014	0.683	.544	.677	.674	.901	.456	100.0	1.000
100	200	0.228	-0.008	0.008	0.012	-0.002	0.008	0.749	.691	.806	.804	.936	.309	100.0	1.000
200	50	0.221	-0.034	0.018	0.038	-0.012	0.019	0.530	.196	.255	.285	.785	.804	100.0	1.000
200	100	0.227	-0.017	0.010	0.019	-0.005	0.010	0.648	.338	.475	.476	.876	.662	100.0	1.000
200	200	0.228	-0.009	0.006	0.011	-0.002	0.006	0.724	.521	.671	.674	.919	.479	100.0	1.000

Table 12: Monte Carlo results: dynamic unbalanced panel (DGP 4, fill = 80%).

		Bias & RMSE						95% Coverage							
N	T	OLS	IFE	SD	RMSE	BC	BC.RMSE	SE/SD	Std	Rob	Cov.HAC	BC.HAC	Size	$P(\hat{r} = r)$	Conv.
30	50	0.222	-0.031	0.042	0.052	-0.015	0.043	0.660	.716	.745	.744	.831	.285	91.6	1.000
30	100	0.224	-0.015	0.025	0.029	-0.005	0.025	0.754	.802	.854	.852	.894	.198	97.3	1.000
30	200	0.227	-0.007	0.018	0.019	-0.002	0.018	0.754	.821	.874	.870	.906	.179	98.2	1.000
50	50	0.223	-0.030	0.031	0.043	-0.015	0.033	0.689	.654	.724	.723	.867	.346	97.7	1.000
50	100	0.224	-0.015	0.019	0.025	-0.005	0.020	0.747	.746	.827	.831	.912	.254	99.8	1.000
50	200	0.228	-0.008	0.014	0.016	-0.003	0.014	0.743	.796	.863	.860	.907	.204	100.0	1.000
100	50	0.221	-0.033	0.024	0.041	-0.017	0.027	0.610	.451	.546	.545	.797	.550	99.7	1.000
100	100	0.227	-0.016	0.014	0.022	-0.006	0.015	0.713	.589	.709	.703	.902	.411	100.0	1.000
100	200	0.229	-0.008	0.010	0.013	-0.003	0.010	0.708	.691	.797	.793	.910	.309	100.0	1.000
200	50	0.223	-0.034	0.018	0.039	-0.017	0.024	0.564	.233	.315	.319	.688	.767	99.8	1.000
200	100	0.225	-0.016	0.011	0.019	-0.006	0.011	0.674	.406	.549	.546	.880	.594	100.0	1.000
200	200	0.227	-0.008	0.007	0.011	-0.003	0.007	0.734	.565	.728	.725	.933	.435	100.0	1.000

Table 13: Monte Carlo results: dynamic unbalanced panel (DGP 4, fill = 60%).

		Bias & RMSE						95% Coverage							
N	T	OLS	IFE	SD	RMSE	BC	BC.RMSE	SE/SD	Std	Rob	Cov.HAC	BC.HAC	Size	$P(\hat{r} = r)$	Conv.
30	50	0.223	-0.025	0.048	0.054	-0.012	0.048	0.655	.751	.748	.752	.809	.250	84.3	0.999
30	100	0.224	-0.012	0.030	0.032	-0.004	0.030	0.740	.836	.866	.865	.901	.164	92.9	1.000
30	200	0.227	-0.006	0.021	0.022	-0.001	0.021	0.746	.840	.875	.872	.884	.160	97.5	1.000
50	50	0.222	-0.026	0.036	0.045	-0.013	0.037	0.666	.720	.755	.754	.833	.281	94.2	1.000
50	100	0.224	-0.014	0.023	0.026	-0.005	0.023	0.739	.784	.829	.825	.895	.216	99.3	1.000
50	200	0.228	-0.007	0.016	0.017	-0.002	0.016	0.752	.819	.875	.870	.909	.181	99.8	1.000
100	50	0.221	-0.030	0.027	0.040	-0.016	0.030	0.630	.549	.629	.627	.807	.451	98.9	1.000
100	100	0.227	-0.015	0.016	0.022	-0.006	0.017	0.710	.667	.763	.765	.885	.333	100.0	1.000
100	200	0.229	-0.008	0.011	0.014	-0.002	0.011	0.720	.759	.849	.848	.914	.241	100.0	1.000
200	50	0.223	-0.031	0.020	0.037	-0.017	0.025	0.588	.349	.439	.440	.724	.651	99.5	1.000
200	100	0.225	-0.016	0.012	0.020	-0.006	0.013	0.676	.519	.645	.646	.873	.482	100.0	1.000
200	200	0.227	-0.008	0.008	0.011	-0.003	0.008	0.734	.640	.784	.778	.924	.360	100.0	1.000

Table 14: Factor selection accuracy across all DGPs ($r_{\text{true}} = 2$, $r_{\text{max}} = 5$).

DGP	Fill	N	T	$\bar{r}$	$P(\hat{r} = 2)\%$	$P(\hat{r} < 2)\%$	$P(\hat{r} > 2)\%$
1	80%	30	10	1.01	0.6	99.4	0.0
1	80%	30	50	1.65	65.1	34.9	0.0
1	80%	30	100	1.88	87.7	12.3	0.0
1	80%	30	200	1.93	92.6	7.4	0.0
1	80%	50	10	1.01	0.8	99.2	0.0
1	80%	50	50	1.98	97.6	2.4	0.0
1	80%	50	100	2.00	100.0	0.0	0.0
1	80%	50	200	2.00	100.0	0.0	0.0
1	80%	100	10	1.00	0.4	99.6	0.0
1	80%	100	50	2.00	100.0	0.0	0.0
1	80%	100	100	2.00	100.0	0.0	0.0
1	80%	100	200	2.00	100.0	0.0	0.0
1	80%	200	10	1.00	0.0	100.0	0.0
1	80%	200	50	2.00	100.0	0.0	0.0
1	80%	200	100	2.00	100.0	0.0	0.0
1	80%	200	200	2.00	100.0	0.0	0.0
2	60%	30	10	1.63	54.1	41.5	4.4
2	60%	30	50	1.75	73.5	26.0	0.5
2	60%	30	100	1.81	80.8	19.2	0.0
2	60%	30	200	1.81	80.9	19.1	0.0
2	60%	50	10	1.71	65.4	31.9	2.7
2	60%	50	50	1.88	87.6	12.4	0.0
2	60%	50	100	1.94	93.7	6.3	0.0
2	60%	50	200	1.96	96.2	3.8	0.0
2	60%	100	10	1.71	69.5	29.8	0.7
2	60%	100	50	1.94	94.0	6.0	0.0
2	60%	100	100	2.00	99.9	0.1	0.0
2	60%	100	200	2.00	100.0	0.0	0.0
2	60%	200	10	1.77	77.2	22.8	0.0
2	60%	200	50	1.97	97.0	3.0	0.0
2	60%	200	100	2.00	99.9	0.1	0.0
2	60%	200	200	2.00	100.0	0.0	0.0
2	80%	30	10	1.71	56.2	37.3	6.5
2	80%	30	50	1.89	83.7	14.6	1.7
2	80%	30	100	1.94	87.6	10.8	1.6
2	80%	30	200	1.95	90.2	8.6	1.2
2	80%	50	10	1.75	69.4	27.9	2.6
2	80%	50	50	1.99	94.9	4.0	1.1
2	80%	50	100	2.00	97.2	2.0	0.8
2	80%	50	200	2.02	98.7	0.5	0.8
2	80%	100	10	1.79	76.0	22.4	1.6
2	80%	100	50	1.99	98.3	1.6	0.1
2	80%	100	100	2.00	99.7	0.1	0.2
2	80%	100	200	2.01	99.6	0.0	0.4
2	80%	200	10	1.81	80.9	18.8	0.3
2	80%	200	50	2.00	99.3	0.6	0.1
2	80%	200	100	2.00	100.0	0.0	0.0
2	80%	200	200	2.00	100.0	0.0	0.0
3	80%	30	50	1.91	90.6	9.4	0.0
3	80%	30	100	1.98	98.2	1.8	0.0
3	80%	30	200	1.99	98.7	1.3	0.0
3	80%	50	50	1.99	99.1	0.9	0.0
3	80%	50	100	2.00	100.0	0.0	0.0
3	80%	50	200	2.00	100.0	0.0	0.0
3	80%	100	50	2.00	100.0	0.0	0.0
3	80%	100	100	2.00	100.0	0.0	0.0
3	80%	100	200	2.00	100.0	0.0	0.0